Patrick McMillan, PhD

Curriculum Vitae

Guelph, Ontario, Canada

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Research Interests

My research interests are primarily in the application of statistics, interpretable machine learning, and novel AI models to spatiotemporal, genomic, and environmental data. I am particularly interested in the use of transformer based models to extract insights from omics data to explain the underlying mechanisms of biological systems. My current work includes developing transformer models for multivariate time series analysis with irregular observations, self-supervised learning for satellite image time series, elucidating the genotype-by-environment interaction for genomic prediction, and applied machine learning for agricultural, environmental monitoring, and animal health data.

Education

PhD, Bioinformatics — University of Guelph
2021 – 2026

- Thesis: *Interpretable Machine Learning for the Modelling of Spatiotemporal Crop Data*
- Advisors: Dr. Lewis Lukens and Dr. Zeny Feng. Committee: Dr. Helen Booker and Dr. Flavio Schenkel.

M.Sc., Applied Statistics — University of Guelph
2019 – 2021

- Thesis: *Regularized Regression Methods and Neural Networks for Modeling Fish Population Health with Water Quality Variables in the Athabasca Oil Sands Region*
- Advisors: Dr. Zeny Feng and Dr. L. Deeth.

BASc, Mathematics and Statistics — University of Guelph
2014 – 2019

Publications

Peer-Reviewed Journal Articles

McMillan, P.G., Feng, Z.Z., Arciszewski, T.J., Proner, R., Deeth, L.E. (2024). Assessing the interpretability–performance trade-off of artificial neural networks using sentinel fish health data. *Environments*, 11(5), 94. doi:10.3390/environments11050094

McMillan, P.G., Feng, Z.Z., Deeth, L.E., Arciszewski, T.J. (2022). Improving monitoring of fish health in the oil sands region using regularization techniques and water quality variables. *Science of the Total Environment*, 811, 152301. doi:10.1016/j.scitotenv.2021.152301

Manuscripts In Revision or Under Review

McMillan, P.G., Lukens, L., Feng, Z.Z. Integrating genomic data and machine learning to improve winter wheat variety performance predictions in Ontario, Canada. *Frontiers in Genetics*. In revision.

McMillan, P.G., Feng, Z.Z., Lukens, L. Transformers reveal the temporal dimension of genetic and environment effects on winter wheat yields in multi-environment trials. *Scientific Reports*. Under review.

McMillan, P.G., Feng, Z.Z., Lukens, L. Label-free forecasting as a pretext task for early-season crop identification. *Remote Sensing of Environment*. Under review.

Manuscripts In Preparation

McMillan, P.G., Tulpan, D., Ribeiro, E.S. Assessing the predictive ability of prepartum data across multiple time points for postpartum disease diagnosis. Target journal: *Journal of Dairy Science*.

Other Work

Bhullar, A., **McMillan, P.G.** (2024). Minimizing climate change-induced food insecurity through land suitability predictions and preemptive breeding of climate change resilient crops. *National Student Paper Competition* (Top 5 finalist). Supervisor: Dr. K. Nadeem.

McMillan, P.G. (2026). *Interpretable Machine Learning for the Modelling of Spatiotemporal Crop Data*. PhD thesis, University of Guelph.

McMillan, P.G. (2021). *Regularized Regression Methods and Neural Networks for Modeling Fish Population Health with Water Quality Variables in the Athabasca Oil Sands Region*. M.Sc. thesis, University of Guelph.

Awards, Scholarships, and Honours

Braithwaite Conference Travel Grant, CFE-CMStatistics 2025 (Institutional) *2025*
Braithwaite Conference Travel Grant, TIES 2024 (Institutional) *2024*
Braithwaite Conference Travel Grant, SSC 2024 Annual Meeting (Institutional) *2024*
National Student Paper Competition, Top 5 Finalist (National) *2024*

- Paper ranked in the top five in Canada. Invited to present to a panel of senior government officials.

Major General LaFleche Memorial Scholarship (Institutional) *2024*
Queen Elizabeth II Graduate Scholarship in Science and Technology (Provincial) *2023*
Sue Chase and John Steckle Graduate Scholarship in Agriculture (Institutional) *2023*
Major General LaFleche Memorial Scholarship (Institutional, $1,000) *2022*
Ontario Graduate Scholarship (Provincial) *2022*
Marian Brennan and Hedley Harrison Memorial Scholarship (Institutional, $1,000) *2022*
Corteva Agriscience New Frontiers: AI in Agriculture Scholarship (International) *2021*

- One of six graduate students selected globally for applying BART to predict cultivar performance in untested environments.

Bayer Crop Science Graduate Mentorship Program *2021*

- One of seven University of Guelph graduate students selected and paired with a Bayer scientist as a mentor.

Research Funding Contributions

Mitacs Accelerate and NSERC Alliance Joint Grant (Awarded) University of Guelph *2025*

- *Development of Streamlined Methods for Assessment of Transition Management Quality and Data-Driven Decisions.* PI: Dr. Eduardo de Souza Ribeiro. Partners: Intervet Canada Corp. (Merck Animal Health), Dairy Farmers of Ontario, Jefo Nutrition, and Lactanet. Contributed to the proposal.

NSERC Grant Application, Co-applicant (Not funded) University of Guelph *2025*

- *Using Deep Learning and Satellite Imagery for Long-Term Agricultural Land Suitability Assessment in Northern Ontario.* PI: Dr. Khurram Nadeem.

Presentations

Invited Talks

McMillan, P.G. (2026, October 7). Elucidating the temporal dimension of the genotype-by-environment interaction with transformers for genomic prediction of crop yields. Department of Computer Science, Western University. London, ON.

McMillan, P.G., Smith, A., Feng, Z.Z., Lukens, L. (2026, June 10). Evaluating the use of genomic information to predict crop field performance. Next-Generation Genomics for Sustainability and Planetary Health, geXc (The Genomic Exchange Community). University of Guelph, Guelph, ON. Only student among the invited speakers.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2025, January 20). Using climate data to provide personalized variety recommendations for Ontario winter wheat farmers. Ontario Cereal Crops Committee Research Day. Guelph, ON.

Bhullar, A., **McMillan, P.G.** (2024, April 11). Data-driven policy building to promote resilient agriculture in a changing climate. National Student Paper Competition finalist presentation to senior government officials. Virtual.

McMillan, P.G. (2023, October 23). Increasing crop yields in Ontario: Two ways of improving the cultivar recommendation process. Bayer Crop Science. Virtual.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2022, January 26). An interactive visualization of crop performance in Ontario. Ontario Cereal Crops Committee Annual Meeting. Virtual.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2021, September 2). Creating a robust and accurate multi-environment genomic prediction model of winter wheat yield using AI. Corteva Agriscience New Frontiers: Artificial Intelligence in Agriculture (speaker and panellist). Virtual.

Contributed Talks

McMillan, P.G., Feng, Z.Z., Lukens, L. (2025, December). Elucidating the temporal dimension of the genotype by environment interaction effect with transformers. 19th International Joint Conference CFE-CMStatistics 2025. Birkbeck, University of London, London, UK.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2024, December 2). Improving crop variety recommendations for farmers: An integrated approach using machine learning, genetics, and climate data. 31st Annual Conference of The International Environmetrics Society (TIES 2024). Adelaide, South Australia, Australia.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2024, June 5). Improving crop variety recommendations for farmers: An integrated approach using machine learning and genetics. Statistical Society of Canada Annual Meeting. St. John's, NL.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2023, May). Genomic prediction of newly developed crop varieties using transformers. Statistical Society of Canada 50th Annual Meeting. Carleton University, Ottawa, ON.

McMillan, P.G., Feng, Z.Z., Deeth, L.E., Arciszewski, T.J. (2021, June). Improving monitoring of fish health in the Athabasca oil sands region using regularization techniques and water quality variables. Statistical Society of Canada Annual Meeting. Virtual.

McMillan, P.G., Feng, Z.Z., Deeth, L.E., Arciszewski, T.J. (2021, June). Improving monitoring of fish health in the Athabasca oil sands region using regularization techniques and water quality variables. Society of Environmental Toxicology and Chemistry Annual General Meeting. Virtual.

McMillan, P.G., Feng, Z.Z., Deeth, L.E., Arciszewski, T.J. (2020). Using nested mixed effect models to predict the health of the aquatic ecosystem found in the Athabasca River in northern Alberta. Statistical Society of Canada Annual Meeting. Virtual.

Posters and Competitions

Cooper, A., **McMillan, P.G.**, Ward, M. (2024, June 5). A joint prediction framework for identifying long and short lengths of stay in the ICU. Case Studies in Data Analysis Competition, Statistical Society of Canada Annual Meeting. St. John's, NL.

McMillan, P.G., Feng, Z.Z., Lukens, L. (2021, July 30). Improving scientific outreach to farmers in Ontario: An interactive application for visualizing winter wheat trial data. Poster, National Association of Plant Breeders Annual Meeting. Virtual.

Research Experience

Doctoral Researcher — University of Guelph
2021 – 2026

- **Genomic prediction:** Compared BLUP and gBLUP with Bayesian additive regression trees (BART) and genomic BART on Ontario winter wheat multi-environment trials. Pairing BART with genomic data reduced mean square prediction error by up to 21%. Interpretability analysis found that predictive markers clustered near known yield loci on chromosomes 1B, 2B, and 3B.
- **Temporal genotype-by-environment effects:** Encoded SNP genotypes as time-varying inputs to a temporal fusion transformer alongside climate time series, with a concrete autoencoder selecting the 25 most informative SNPs. The model reached a hold-out R^2 of 0.519 against 0.430 for gBLUP and identified when in the season genotype, environment, and their interaction mattered most.
- **Early-season crop identification:** Trained a label-free forecaster on Harmonized Landsat Sentinel-2 time series from over 1,000,000 fields to complete each field's early-season spectral curves. Adding the forecaster raised crop classification accuracy one month into the season from 57.5% to 78.7% with far fewer labels than conventional methods.
- Trained models on Digital Research Alliance of Canada clusters with multi-GPU (H100) pipelines.

Graduate Research Assistant — University of Guelph
Sep 2024 – Sep 2025

- Statistician on a team predicting postpartum disease in dairy cows from records and collar sensor data collected 60, 21, and 1 day before calving. Supervisor: Dr. Eduardo Ribeiro.
- Built penalised regression and machine learning classifiers that reached 75% accuracy, higher than previously published work. Manuscript in preparation.

Data Analyst — Agri-Food Data Canada
Jun 2022 – Present

- Supervisor: Dr. Michelle Edwards.
- Statistical consulting on predictive modelling, imputation, and data analysis for other departments.
- Built a PostgreSQL database holding over 50 years of data from long-term experiments, updated in real time as new data arrive.
- Developed and contributed to three research data portals (see Software and Data Tools).

M.Sc. Researcher — University of Guelph
2019 – 2021

- Modelled trout-perch health in the Athabasca oil sands region from water quality data. Used penalised regression to identify the most influential contaminants and neural networks to predict fish health.
- Identified two influential metals that existing monitoring programs did not track. Published two journal articles.

Undergraduate Researcher University of Guelph
2019

- Supervisor: Dr. Julie Horrocks. With the University of Guelph Alumni Association, applied classification models to records of over 300,000 graduates and identified more than 200 prospective major donors.

Teaching Experience

Guest Lecturer, STAT*3510 Environmental Risk Assessment University of Guelph
Winter 2026

- One-week lecture series on bioassay and dose-response modelling.

Teaching Assistant University of Guelph
2019 – 2026

- Over 20 course offerings, often as head TA. Courses include STAT*3210 Experimental Design, STAT*3510 Environmental Risk Assessment, and STAT*2120 Probability and Statistics for Engineers.

Lead Instructor, Statistical Methods for High-Dimensional Data Bioinformatics.ca
Aug 2024

- Led the team that developed and taught this three-day national workshop. One of ten workshops selected that year, and the only one led by graduate students.

Instructor, Statistical Modelling and Programming Course Environment and Climate Change Canada
Jan 2024

- Co-developed and taught a three-day course for federal researchers at the Canada Centre for Inland Waters, Burlington, ON.

Guest Lecturer, STAT*6920 Statistical Computing University of Guelph
2019

- One-week lecture series on data wrangling and visualisation in R (tidyverse), with an assignment. The materials are still used in the course.

Mentorship and Supervision

Co-op Student Supervisor LandWise Analytica
May – Sep 2025

Industry Advisor, Big Data Analytics Capstone Georgian College
May – Dec 2024

- Scoped and advised a five-student master's project on tactics gambling operators use to encourage problem gambling.

Co-op Student Supervisor LandWise Analytica
May – Sep 2024

Service and Professional Memberships

Hiring Committee Member, Department of Mathematics and Statistics University of Guelph
2024

Hiring Committee Member, Department of Mathematics and Statistics University of Guelph
2023

Hiring Committee Member, Department of Mathematics and Statistics University of Guelph
2023

Member Statistical Society of Canada
2021 – 2025

Member The International Environmetrics Society
2024

Industry and Consulting Experience

Co-Founder Statistical and AI Consultancy
2023 – Present

- Statistical and AI consulting for academic, government, and industry clients, including The Blood Project (Harvard), Integram, and Ontario's chicken and turkey processor associations.

Co-Founder LandWise Analytica
Dec 2022 – Jan 2026

- AI AgriTech startup automating farmland due diligence with geospatial and climate data.
- Secured investment led by NEC X!, the only company of 350 applicants offered investment in the final phase. Peak valuation of $5M USD.
- Pitched at the World Agri-Tech Summit (San Francisco) and the Agri-Tech Venture Forum (Toronto).

Project Assistant Media Sonar
Jan – Mar 2020

- Contributed to a machine learning model that flags social media posts promoting violence for law enforcement partners.

Software and Data Tools

- **Long-Term Cropping Trial Data Portal (Agri-Food Data Canada):** Results from long-term trials at the Elora Research Station. agrifooddatacanada.ca/research-centre-data-access
- **Ontario Dairy Research Centre Data Portal (Agri-Food Data Canada):** Live data from dairy cows at the research centre. odrcdata.uoguelph.ca
- **Cover Crop Data Portal (Agri-Food Data Canada):** Not yet public.
- **Winter Wheat Performance App (R Shiny):** Ontario variety trial results by region, developed for OMAFRA. pmcmillan.shinyapps.io
- **Open-source analysis code:** Model and hyperparameter optimisation code for the genomic prediction study. github.com/patmcmilla

Technical Skills

- **Machine Learning and AI:** Transformers (TFT), CNNs, LSTMs, autoencoders, self-supervised and pretext learning, BART, random forests, XGBoost, SHAP, hyperparameter optimisation.
- **Statistics:** Penalised regression (LASSO, elastic net), generalised linear models, linear mixed models (BLUP, gBLUP), missing data imputation, clustering (hierarchical, k-means, DBSCAN), dimension reduction (PCA, PCoA), longitudinal and spatiotemporal data, experimental design, survival and risk modelling.
- **Genomics and Bioinformatics:** Genomic prediction (gBLUP, RKHS, BART), genome-wide association studies (GWAS), QTL and candidate gene interpretation, SNP genotype and RNA-seq gene expression data. Packages: BGLR, rrBLUP, AGHmatrix, statgenGWAS, vcfR.
- **Remote Sensing:** Harmonized Landsat Sentinel-2 imagery, Google Earth Engine (Daymet, MODIS, OpenLandMap, SRTM).
- **Computing:** R, Python, SQL (PostgreSQL), PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, Hugging Face, Git, Docker, Slurm on HPC clusters, R Shiny, LaTeX.

References

- Dr. Zeny Feng, Department of Mathematics and Statistics, University of Guelph. zfeng@uoguelph.ca (PhD and M.Sc. co-advisor)
- Dr. Lewis Lukens, Department of Plant Agriculture, University of Guelph. llukens@uoguelph.ca (PhD co-advisor)
- Dr. Michelle Edwards, Agri-Food Data Canada, University of Guelph. edwardsm@uoguelph.ca (supervisor, Data Analyst position)